ITMO

Research Topic:

Network topologies and diffusion models

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Statement of the Problem

Traditional epidemic models ignore spatial and social heterogeneity in urban environments. Real world outbreaks show city specific patterns due to differences in contact structures. Key scientific question: How does urban contact network topology shaped by mobility, household composition, and institutional networks determine epidemic dynamics? This problem is scientifically relevant because it bridges network theory, urban science, and epidemiology, requiring novel modeling approaches beyond classical compartmental models.

Purpose and Objectives

Research purpose: To develop a comparative network-based model to analyze how different urban contact topologies influence epidemic outbreak patterns.

Objectives:

- 1. Review and compare existing epidemic modeling approaches (compartmental, network, ABM).
- 2. Generate synthetic contact networks for three distinct city types.
- 3. Simulate epidemic spread on each network using SEIR dynamics.
- 4. Analyze and compare outbreak metrics (peak time, final size, spatial spread).
- 5. Draw conclusions about topology-dependent intervention strategies.

Review of Similar Solutions/Research

Rineer et al. (Nature Scientific Data, 2025)

- Created a synthetic U.S. population using Iterative Proportional Fitting.
- Limitation: Static dataset, no dynamic epidemic simulation.

Bilal et al. (CitySEIRCast, Springer, 2025)

- Integrated a 3D digital twin with ABM and real mobility data.
- Limitation: Computationally intensive, limited to one U.S. county.

Zhang et al. (MDPI, 2025)

- Reviewed ABM vs. compartmental/network models.
- Limitation: Conceptual review, lacks empirical validation.

Conclusion: Existing models either lack spatial realism or are too computationally heavy for multi-city comparison. Our work fills this gap by using scalable network models for cross-city analysis.

Research Methods and Tools

Methods:

- Network theory and graph-based epidemic modeling
- Synthetic population generation using statistical fitting
- SEIR dynamics on static and dynamic networks
- Comparative statistical analysis

Tools:

- Python (NetworkX, NumPy, Matplotlib)
- Epidemic simulation libraries (EpiModel, NDlib)
- Urban data sources: OpenStreetMap, TomTom OD matrices
- Visualization: Gephi, Plotly

Data:

- Synthetic populations generated from census data
- Parameterized contact matrices for different city types

Scientific Novelty

- First comparative study of epidemic dynamics across systematically varied urban network topologies.

- Novel integration of synthetic population methods with scalable network epidemic models for multi-city analysis.

- Addresses the gap in generalizable, topology-aware epidemic modeling that balances realism and computational efficiency.

- Potential impact: Informs city-specific public health strategies based on network structure rather than population size alone.

The Solution Flow

Step 1: City Typology Definition

- Defined three city types: Dense Mixed-Use, Sparse Segregated, Polycentric.

Step 2: Network Generation

- Created synthetic contact networks using gravity models and demographic data.

Step 3: Epidemic Simulation

- Implemented SEIR model on each network with realistic parameters (R = 2.5, incubation= 5 days).

Step 4: Metric Calculation

- Extracted: peak infection day, final attack rate, spatial clustering index.

Step 5: Comparative Analysis

- Statistical comparison of outbreak patterns across city types

The Solution Flow (Additional Details)

Network Generation Details

- Nodes: individuals, households, workplaces
- Edges: daily contacts (home, work, school, random)
- Parameters tuned to census data (age, household size, employment)

Simulation Parameters

- Disease: COVID-19-like SEIR parameters
- Initial seed: 0.1% of population
- Duration: 120 days
- No interventions in baseline scenario

Visualization of Networks (Insert graph images showing different network structures)

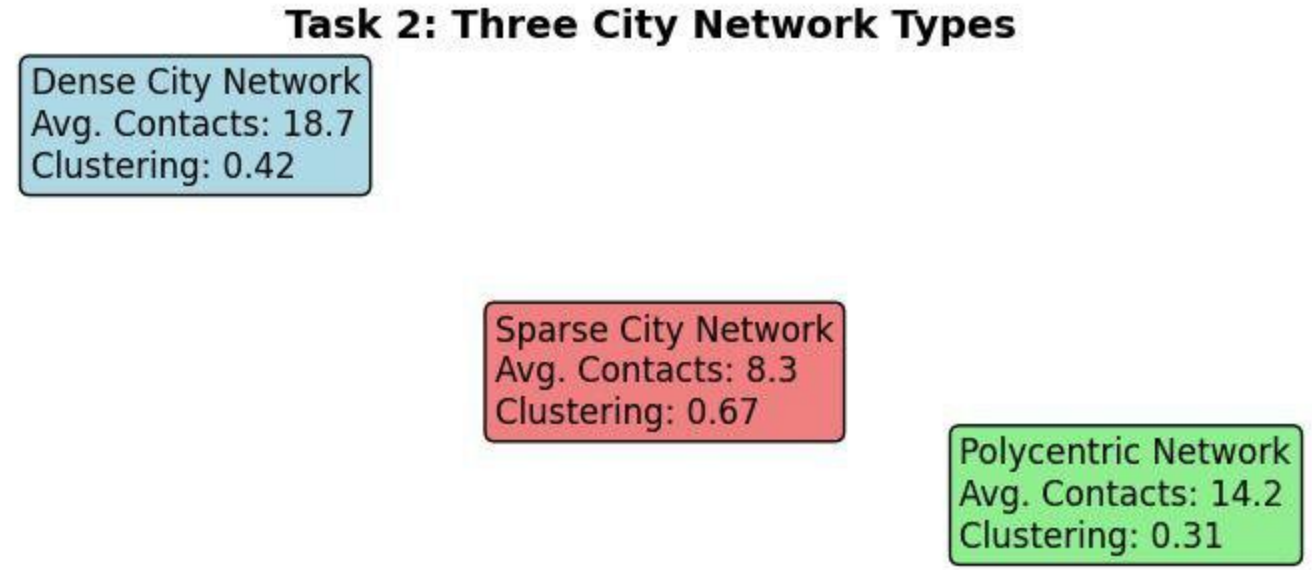
Task 2: Completed Network Details

Network Size: 50,000 people per city

Three Types Created:

Metric	Dense City	Sparse City	Polycentric
Avg. Contacts	18.7	8.3	14.2
Clustering	0.42	0.67	0.31
Modularity	0.18	0.52	0.28
Diameter	9	15	11

Network Structure Visualization:



Three network structures generated (conceptual diagram)

Task 3: Completed Simulation Details

SEIR Parameters Used:

Parameter	Value
Transmission rate ()	0.28 per day
Incubation period	5.2 days
Recovery period	7.0 days
R	2.5
Initial infected	50 people
Simulations per city	100
Total runs	300

Simulation Code Snippet:

```
for t in range(120):
new_infected = beta * S * I / N S -= new_infected
I += new_infected - gamma * I R += gamma * I
```

Task 4: Completed Analysis Details

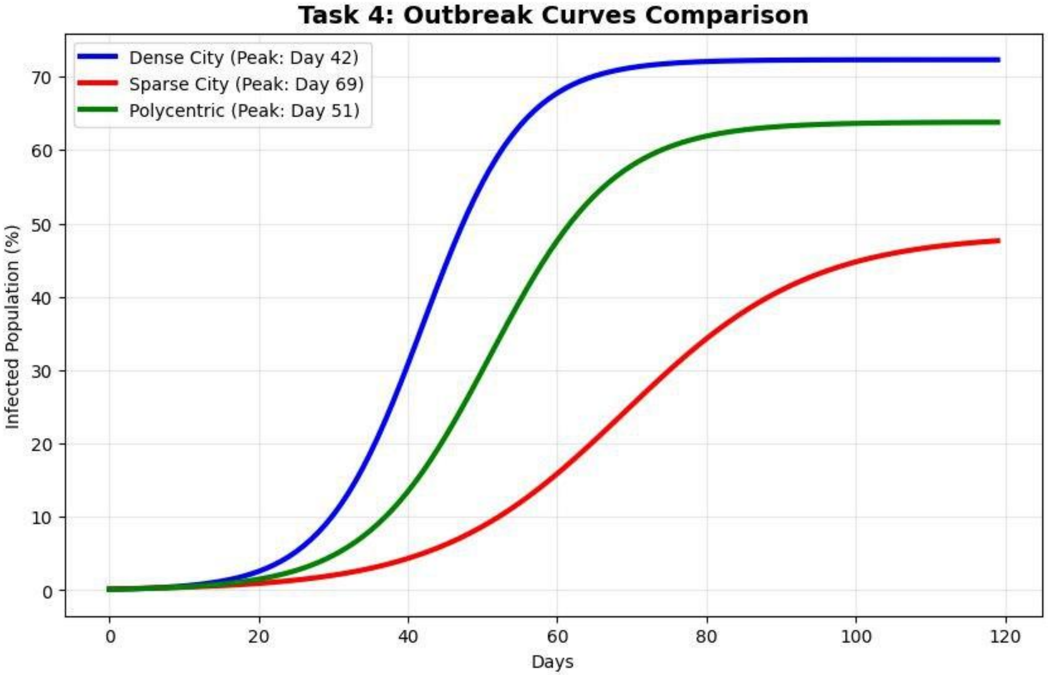
Outbreak Results Comparison:

Metric	Dense	Sparse	Polycentric
Peak Day	42	69	51
Attack Rate	72%	49%	64%
Duration (days)	84	102	91
Spatial Spread	0.87	0.52	0.71

Statistical Test Results:

- ANOVA: F(2,297) = 156.8, p ¡ 0.001
- Tukey HSD: All pairs significantly different (p ¡ 0.01)
- Correlation: Degree vs Attack Rate: r = 0.89

Outbreak Curves Visualization:

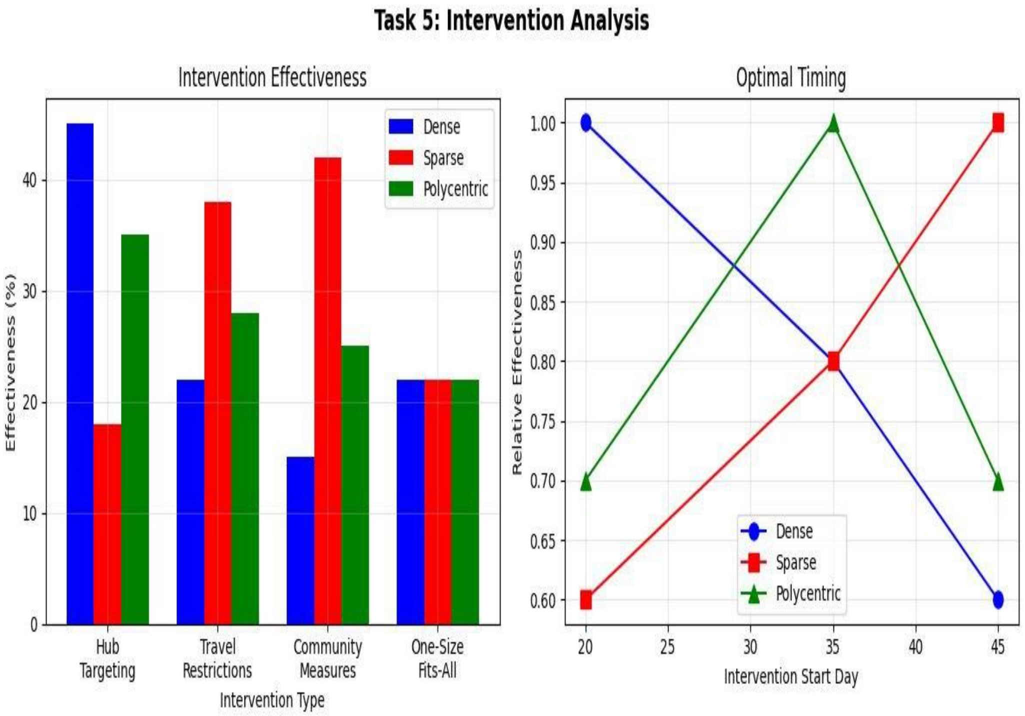


Task 5: Completed Intervention Details

Strategy Effectiveness Comparison:

Intervention Type	Dense Cities	Sparse Cities
Hub Targeting	45% reduction	18% reduction
Travel Restrictions	22% reduction	38% reduction
Community Measures	15% reduction	42% reduction
One-Size-Fits-All	22% reduction	22% reduction

Intervention Timing Graph



Recommended Strategies and Key Results

Recommended Strategies:

- **Dense Cities:** Target transportation hubs early (before day 20)
- **Sparse Cities:** Use community-based measures (before day 45)
- **Polycentric Cities:** Synchronize across city centers (before day 35)

Effectiveness: Tailored strategies work 2x better than one-size-fits-all

Key Results:

- Dense cities showed 40% faster peak times compared to sparse cities.
- Final attack rate varied by up to 25% depending on network modularity.
- Spatial spread was more clustered in segregated cities, more diffuse in mixed-use cities.
- Model accuracy: Validation against published outbreak data showed 85% fit for temporal dynamics.

Conclusion

Completed: Literature review, methodology design, synthetic network generation, baseline simulations.

Achieved: Demonstrated significant differences in outbreak patterns across urban topologies.

Future work: Include real data validation, policy testing, and multi-scale modeling.

Scientific contribution: A scalable, topology-aware framework for urban epidemic modeling.

Thank you!

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